In [1]:
```
import torch
import numpy as np
import random

random.seed(42)
np.random.seed(42)
torch.manual_seed(42)
```

Out[1]:
```
<torch._C.Generator at 0x10ff123f0>
```

In [2]:
```
import warnings

warnings.filterwarnings("ignore")
```

In [3]:
```
import sys
!{sys.executable} -m pip install torch "gluonts[torch]"
```

```
Requirement already satisfied: torch in /opt/miniconda3/envs/econ491_walmart
_forcasting/lib/python3.10/site-packages (2.11.0)
Requirement already satisfied: gluonts[torch] in /opt/miniconda3/envs/econ49
1_walmart_forcasting/lib/python3.10/site-packages (0.16.2)
Requirement already satisfied: filelock in /opt/miniconda3/envs/econ491_walm
art_forcasting/lib/python3.10/site-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /opt/miniconda3/
envs/econ491_walmart_forcasting/lib/python3.10/site-packages (from torch)
(4.15.0)
Requirement already satisfied: setuptools<82 in /opt/miniconda3/envs/econ491
_walmart_forcasting/lib/python3.10/site-packages (from torch) (81.0.0)
Requirement already satisfied: sympy>=1.13.3 in /opt/miniconda3/envs/econ491
_walmart_forcasting/lib/python3.10/site-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /opt/miniconda3/envs/econ4
91_walmart_forcasting/lib/python3.10/site-packages (from torch) (3.4.2)
Requirement already satisfied: jinja2 in /opt/miniconda3/envs/econ491_walmar
t_forcasting/lib/python3.10/site-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /opt/miniconda3/envs/econ491
_walmart_forcasting/lib/python3.10/site-packages (from torch) (2025.12.0)
Requirement already satisfied: numpy<2.2,>=1.16 in /opt/miniconda3/envs/econ
491_walmart_forcasting/lib/python3.10/site-packages (from gluonts[torch])
(2.1.3)
Requirement already satisfied: pandas<3,>=1.0 in /opt/miniconda3/envs/econ49
1_walmart_forcasting/lib/python3.10/site-packages (from gluonts[torch])
(2.3.3)
Requirement already satisfied: pydantic<3,>=1.7 in /opt/miniconda3/envs/econ
491_walmart_forcasting/lib/python3.10/site-packages (from gluonts[torch])
(2.12.5)
Requirement already satisfied: tqdm~=4.23 in /opt/miniconda3/envs/econ491_wa
lmart_forcasting/lib/python3.10/site-packages (from gluonts[torch]) (4.67.3)
Requirement already satisfied: toolz~=0.10 in /opt/miniconda3/envs/econ491_w
almart_forcasting/lib/python3.10/site-packages (from gluonts[torch]) (0.1
2.1)
Requirement already satisfied: lightning<2.5,>=2.2.2 in /opt/miniconda3/env
s/econ491_walmart_forcasting/lib/python3.10/site-packages (from gluonts[torc
h]) (2.4.0)
Requirement already satisfied: pytorch-lightning<2.5,>=2.2.2 in /opt/minicon
da3/envs/econ491_walmart_forcasting/lib/python3.10/site-packages (from gluon
ts[torch]) (2.4.0)
Requirement already satisfied: scipy<1.16.0 in /opt/miniconda3/envs/econ491_
walmart_forcasting/lib/python3.10/site-packages (from gluonts[torch]) (1.1
5.3)
Requirement already satisfied: PyYAML<8.0,>=5.4 in /opt/miniconda3/envs/econ
491_walmart_forcasting/lib/python3.10/site-packages (from lightning<2.5,>=
2.2.2->gluonts[torch]) (6.0.3)
Requirement already satisfied: lightning-utilities<2.0,>=0.10.0 in /opt/mini
conda3/envs/econ491_walmart_forcasting/lib/python3.10/site-packages (from li
ghtning<2.5,>=2.2.2->gluonts[torch]) (0.15.3)
Requirement already satisfied: packaging<25.0,>=20.0 in /opt/miniconda3/env
s/econ491_walmart_forcasting/lib/python3.10/site-packages (from lightning<
2.5,>=2.2.2->gluonts[torch]) (24.2)
Requirement already satisfied: torchmetrics<3.0,>=0.7.0 in /opt/miniconda3/e
nvs/econ491_walmart_forcasting/lib/python3.10/site-packages (from lightning<
2.5,>=2.2.2->gluonts[torch]) (1.9.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /opt/miniconda
3/envs/econ491_walmart_forcasting/lib/python3.10/site-packages (from fsspec
```

```
[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts[torch]) (3.13.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/miniconda3/env
s/econ491_walmart_forcasting/lib/python3.10/site-packages (from pandas<3,>=
1.0->gluonts[torch]) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /opt/miniconda3/envs/econ491_
walmart_forcasting/lib/python3.10/site-packages (from pandas<3,>=1.0->gluont
s[torch]) (2026.1.post1)
Requirement already satisfied: tzdata>=2022.7 in /opt/miniconda3/envs/econ49
1_walmart_forcasting/lib/python3.10/site-packages (from pandas<3,>=1.0->gluo
nts[torch]) (2025.3)
Requirement already satisfied: annotated-types>=0.6.0 in /opt/miniconda3/env
s/econ491_walmart_forcasting/lib/python3.10/site-packages (from pydantic<3,>
=1.7->gluonts[torch]) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in /opt/miniconda3/env
s/econ491_walmart_forcasting/lib/python3.10/site-packages (from pydantic<3,>
=1.7->gluonts[torch]) (2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in /opt/miniconda3/e
nvs/econ491_walmart_forcasting/lib/python3.10/site-packages (from pydantic<
3,>=1.7->gluonts[torch]) (0.4.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /opt/miniconda3/en
vs/econ491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=
4.0.0a0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gl
uonts[torch]) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in /opt/miniconda3/envs/econ
491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts[tor
ch]) (1.4.0)
Requirement already satisfied: async-timeout<6.0,>=4.0 in /opt/miniconda3/en
vs/econ491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=
4.0.0a0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gl
uonts[torch]) (5.0.1)
Requirement already satisfied: attrs>=17.3.0 in /opt/miniconda3/envs/econ491
_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a0,!=
4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts[torc
h]) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /opt/miniconda3/envs/eco
n491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a
0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts
[torch]) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /opt/miniconda3/envs/e
con491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a
0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts
[torch]) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in /opt/miniconda3/envs/econ
491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a0,!
=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts[tor
ch]) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /opt/miniconda3/envs/eco
n491_walmart_forcasting/lib/python3.10/site-packages (from aiohttp!=4.0.0a
0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2->gluonts
[torch]) (1.23.0)
Requirement already satisfied: idna>=2.0 in /opt/miniconda3/envs/econ491_wal
mart_forcasting/lib/python3.10/site-packages (from yarl<2.0,>=1.17.0->aiohtt
p!=4.0.0a0,!=4.0.0a1->fsspec[http]<2026.0,>=2022.5.0->lightning<2.5,>=2.2.2-
>gluonts[torch]) (3.11)
Requirement already satisfied: six>=1.5 in /opt/miniconda3/envs/econ491_walm
```

```
art_forcasting/lib/python3.10/site-packages (from python-dateutil>=2.8.2->pa
ndas<3,>=1.0->gluonts[torch]) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/miniconda3/envs/ec
on491_walmart_forcasting/lib/python3.10/site-packages (from sympy>=1.13.3->t
orch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/miniconda3/envs/econ4
91_walmart_forcasting/lib/python3.10/site-packages (from jinja2->torch)
(3.0.2)
```

In [4]:
```
import pandas as pd
import numpy as np
from gluonts.dataset.common import ListDataset
from gluonts.torch.model.deepar import DeepAREstimator
from gluonts.evaluation import make_evaluation_predictions, Evaluator
import statsmodels.formula.api as smf
import gc
import os
```

In [5]:
```
# load data
os.chdir("/Users/conniezhang/Desktop/ECON491/data/")

sales_train = pd.read_csv("Sales Train Validation.csv")
calendar = pd.read_csv("Calendar (1).csv")
prices = pd.read_csv("sell_prices.csv")
```

In [6]:
```
# process the sales data

# Drop unused columns to save RAM immediately
sales_train = sales_train.drop(columns=["cat_id", "state_id"])

# Find all the day columns (d_1, d_2, etc.)
day_cols = [col for col in sales_train.columns if col.startswith('d_')]

# Melt the dataframe (equivalent to your long_sales in R)
long_sales = pd.melt(
    sales_train,
    id_vars=['item_id', 'dept_id', 'store_id'],
    value_vars=day_cols,
    var_name='d',
    value_name='sales'
)

# Clear old wide data from RAM
del sales_train
gc.collect()
```

Out[6]: 80

In [7]:
```
# prepare calendar data and join
calendar['d'] = 'd_' + (calendar.index + 1).astype(str)

# Keep only what we need
slim_calendar = calendar[['d', 'date', 'wm_yr_wk', 'wday', 'month', 'event_t
slim_calendar['date'] = pd.to_datetime(slim_calendar['date'])
```

```
# Merge sales and calendar (like your on="d" join)
final_df = pd.merge(long_sales, slim_calendar, on='d', how='left')

del long_sales, slim_calendar
gc.collect()
```

Out[7]: 0

In [8]:
```
# get average price for each item/store combination
# Get lifetime sales per item for weights
item_weights = final_df.groupby(['item_id', 'store_id', 'dept_id'])['sales']

# Merge prices with weights
prices_mapped = pd.merge(prices, item_weights, on=['item_id', 'store_id'])

# Calculate weighted average price per department per week
prices_mapped['weighted_price'] = prices_mapped['sell_price'] * prices_mappe
dept_prices = prices_mapped.groupby(['store_id', 'dept_id', 'wm_yr_wk']).app
    lambda x: x['weighted_price'].sum() / x['item_lifetime_sales'].sum()
).reset_index(name='avg_price')

del prices, prices_mapped, item_weights
gc.collect()
```

Out[8]: 0

In [9]:
```
# assemble model data
# Aggregate daily sales to store-dept level
model_data = final_df.groupby(['store_id', 'dept_id', 'date', 'wm_yr_wk', 'w

# Join prices
model_data = pd.merge(model_data, dept_prices, on=['store_id', 'dept_id', 'w

# Drop rows with missing prices
model_data = model_data.dropna(subset=['avg_price'])

# Create Dummy Variables for the Econometrics Model
model_data['day_of_week'] = model_data['wday'].astype(str) # Treat as catego
model_data['month_factor'] = model_data['month'].astype(str) # Treat as cate
model_data['is_holiday'] = np.where(model_data['event_type_1'].isna(), 0, 1)
```

In [10]:
```
# train/test split
split_date = pd.to_datetime("2016-04-24")
train_data = model_data[model_data['date'] <= split_date]
test_data = model_data[model_data['date'] > split_date].copy()
```

In [11]:
```
# run the simple benchmark model
# In Python's statsmodels, 'C()' tells it to treat a variable as a categoric
formula = "total_sales ~ avg_price + C(day_of_week) + C(month_factor) + is_h

# Fit the model
simple_model = smf.ols(formula=formula, data=train_data).fit()

# Print the beautiful Econometrics summary table!
print(simple_model.summary())
```

```
                            OLS Regression Results
==============================================================================
==
Dep. Variable:            total_sales   R-squared:                       0.1
23
Model:                            OLS   Adj. R-squared:                  0.1
22
Method:                 Least Squares   F-statistic:                     98
4.3
Date:                Mon, 13 Apr 2026   Prob (F-statistic):
0.00
Time:                        12:07:58   Log-Likelihood:            -1.0384e
+06
No. Observations:              133910   AIC:                         2.077e
+06
Df Residuals:                  133890   BIC:                         2.077e
+06
Df Model:                          19
Covariance Type:            nonrobust
==============================================================================
=============
                             coef    std err          t      P>|t|      [0.02
5      0.975]
------------------------------------------------------------------------------
-------------
Intercept              1263.4202      8.391    150.566      0.000    1246.97
4    1279.867
C(day_of_week)[T.2]      -3.4308      5.793     -0.592      0.554     -14.78
4       7.923
C(day_of_week)[T.3]    -121.9665      5.796    -21.042      0.000    -133.32
7    -110.606
C(day_of_week)[T.4]    -159.1549      5.772    -27.576      0.000    -170.46
7    -147.843
C(day_of_week)[T.5]    -163.9375      5.775    -28.389      0.000    -175.25
6    -152.619
C(day_of_week)[T.6]    -160.5140      5.780    -27.771      0.000    -171.84
2    -149.186
C(day_of_week)[T.7]    -104.8022      5.770    -18.163      0.000    -116.11
1     -93.493
C(month_factor)[T.10]    15.3010      7.628      2.006      0.045       0.35
0      30.252
C(month_factor)[T.11]    -7.6814      7.693     -0.999      0.318     -22.75
9       7.396
C(month_factor)[T.12]   -10.5144      7.630     -1.378      0.168     -25.47
0       4.441
C(month_factor)[T.2]     10.2479      7.465      1.373      0.170      -4.38
4      24.880
C(month_factor)[T.3]      6.0649      7.300      0.831      0.406      -8.24
3      20.373
C(month_factor)[T.4]      4.3948      7.417      0.593      0.553     -10.14
2      18.932
C(month_factor)[T.5]    -20.5718      7.628     -2.697      0.007     -35.52
3      -5.621
C(month_factor)[T.6]     12.3148      7.692      1.601      0.109      -2.76
1      27.391
C(month_factor)[T.7]     17.3144      7.631      2.269      0.023       2.35
```

```
8      32.271
C(month_factor)[T.8]     26.9269      7.638      3.525      0.000      11.95
6      41.898
C(month_factor)[T.9]     22.7449      7.699      2.954      0.003       7.65
6      37.834
avg_price              -209.9684      1.623   -129.372      0.000    -213.14
9    -206.787
is_holiday              -25.5393      5.775     -4.423      0.000     -36.85
8     -14.221
==============================================================================
==
Omnibus:                    51438.959   Durbin-Watson:                   0.0
77
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           240219.6
59
Skew:                           1.840   Prob(JB):
0.00
Kurtosis:                       8.432   Cond. No.                         4
4.2
==============================================================================
==

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is corre
ctly specified.
```

In [12]:

```python
# reshaping data for deepAR
ts_data = model_data.pivot_table(
    index='date',
    columns=['store_id', 'dept_id'],
    values='total_sales',
    aggfunc='sum'
)

# 1. Convert to datetime
ts_data.index = pd.to_datetime(ts_data.index)

# 2. Force daily frequency (this might create rows with NaNs for missing day
ts_data = ts_data.asfreq('D')

# 3. NOW fill everything with 0 (this catches both original missing data AND
ts_data = ts_data.fillna(0)

# Preview the data
ts_data.iloc[:5, :4]
```

Out[12]:

store_id				CA_1
dept_id	FOODS_1	FOODS_2	FOODS_3	HOBBIES_1
date				
2011-01-29	297	674	2268	528
2011-01-30	284	655	2198	489
2011-01-31	214	396	1398	409
2011-02-01	175	476	1607	383
2011-02-02	182	354	1496	263

In [13]:

```
# 1. Get a list of all unique departments from your pivot table
departments = ts_data.columns.get_level_values('dept_id').unique()

# 2. Create dictionaries to store your trained models and datasets
trained_predictors = {}
test_datasets = {}

prediction_length = 28
split_date = pd.to_datetime("2016-04-24")

# 3. Loop through each department and train a separate model
for dept in departments:
    print(f"--- Preparing and Training Model for Department: {dept} ---")

    # Isolate the data just for this department across all stores
    # .xs grabs a "cross-section" of your MultiIndex columns
    dept_ts_data = ts_data.xs(dept, level='dept_id', axis=1)

    # Split train/test for this specific department
    train_dept_ts = dept_ts_data[dept_ts_data.index <= split_date]
    test_dept_ts = dept_ts_data

    # Create GluonTS Datasets
    train_ds = ListDataset(
        [{"start": train_dept_ts.index[0], "target": train_dept_ts[col].valu
        freq="D"
    )

    test_ds = ListDataset(
        [{"start": test_dept_ts.index[0], "target": test_dept_ts[col].values
        freq="D"
    )

    # Initialize the DeepAR model for this department
    estimator = DeepAREstimator(
        freq="D",
        prediction_length=prediction_length,
        context_length=prediction_length * 2,
        num_layers=2,
        hidden_size=32,
        trainer_kwargs={
```

```
            "max_epochs": 15,  # You might want to lower this so the loop do
            "accelerator": "cpu",
            "enable_progress_bar": False
        }
    )

    # Train the model
    predictor = estimator.train(train_ds)

    # Save the trained predictor and test dataset to our dictionaries
    trained_predictors[dept] = predictor
    test_datasets[dept] = test_ds

print("All Department Models Trained Successfully!")
```

```
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
--------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
------------------------------------------------------------------------------
--------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Model for Department: FOODS_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.22353 (best 6.22353), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_25/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.47165 (best 5.47165), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.42801 (best 5.42801), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.34813 (best 5.34813), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.28710 (best 5.28710), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.24473 (best 5.24473), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.18779 (best 5.18779), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' was not in top 1
Epoch 8, global step 450: 'train_loss' reached 5.17996 (best 5.17996), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.17037 (best 5.17037), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
5/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 5.15079 (best 5.15079), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
25/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' reached 5.13631 (best 5.13631), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
25/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' reached 5.12972 (best 5.12972), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
25/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-------------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
------------------------------------------------------------------------------
-------------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
```

```
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Model for Department: FOODS_2 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.66480 (best 6.66480), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_26/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.77745 (best 5.77745), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.56568 (best 5.56568), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.54209 (best 5.54209), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.47958 (best 5.47958), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.47187 (best 5.47187), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' was not in top 1
Epoch 7, global step 400: 'train_loss' reached 5.40499 (best 5.40499), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.39341 (best 5.39341), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.37024 (best 5.37024), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
6/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 5.36967 (best 5.36967), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
26/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' was not in top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' reached 5.36540 (best 5.36540), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
26/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-------------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
------------------------------------------------------------------------------
-------------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
```

```
--- Preparing and Training Model for Department: FOODS_3 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 7.92295 (best 7.92295), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_27/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 7.22134 (best 7.22134), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 6.88381 (best 6.88381), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 6.73197 (best 6.73197), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 6.67545 (best 6.67545), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 6.64568 (best 6.64568), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 6.63355 (best 6.63355), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 6.62940 (best 6.62940), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 6.59971 (best 6.59971), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
7/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' was not in top 1
Epoch 11, global step 600: 'train_loss' reached 6.58713 (best 6.58713), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
27/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' reached 6.55386 (best 6.55386), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
27/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' reached 6.54889 (best 6.54889), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
27/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-------------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
------------------------------------------------------------------------------
-------------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
```

```
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Model for Department: HOBBIES_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.37202 (best 6.37202), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_28/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.60868 (best 5.60868), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.50784 (best 5.50784), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.44897 (best 5.44897), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.42498 (best 5.42498), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.42174 (best 5.42174), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.41818 (best 5.41818), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 5.39849 (best 5.39849), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.39815 (best 5.39815), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.39477 (best 5.39477), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
8/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 5.38060 (best 5.38060), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
28/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' reached 5.38010 (best 5.38010), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
28/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' reached 5.35363 (best 5.35363), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
28/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
--------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
--------------------------------------------------------------------------------
-----------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
```

```
17.6 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Model for Department: HOBBIES_2 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 4.06750 (best 4.06750), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_29/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 3.79151 (best 3.79151), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 3.71615 (best 3.71615), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 3.71147 (best 3.71147), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 3.68137 (best 3.68137), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 3.67184 (best 3.67184), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 3.66955 (best 3.66955), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' was not in top 1
Epoch 8, global step 450: 'train_loss' reached 3.64231 (best 3.64231), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_2
9/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' reached 3.63505 (best 3.63505), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
29/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' was not in top 1
Epoch 13, global step 700: 'train_loss' reached 3.62569 (best 3.62569), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
29/checkpoints/epoch=13-step=700.ckpt' as top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
------------------------------------------------------------------------------
-----------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Model for Department: HOUSEHOLD_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.97734 (best 6.97734), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_30/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 6.16057 (best 6.16057), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.81472 (best 5.81472), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.75054 (best 5.75054), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.70986 (best 5.70986), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.70944 (best 5.70944), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.66998 (best 5.66998), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 5.65092 (best 5.65092), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.63714 (best 5.63714), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.62944 (best 5.62944), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
0/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' was not in top 1
Epoch 11, global step 600: 'train_loss' reached 5.58871 (best 5.58871), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
30/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' was not in top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
--------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.6 K | train | [[1, 1], [1, 1], [1, 1148, 4],
[1, 1148], [1, 1148], [1, 28, 4]] | [1, 100, 28]
--------------------------------------------------------------------------------
-----------------------------------------------------
17.6 K    Trainable params
0         Non-trainable params
17.6 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
```

```
--- Preparing and Training Model for Department: HOUSEHOLD_2 ---
Epoch 0, global step 50: 'train_loss' reached 5.44436 (best 5.44436), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_31/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 4.89822 (best 4.89822), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 4.72725 (best 4.72725), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 4.54978 (best 4.54978), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 4.49688 (best 4.49688), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' was not in top 1
Epoch 6, global step 350: 'train_loss' reached 4.47392 (best 4.47392), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 4.45260 (best 4.45260), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
1/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' was not in top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' reached 4.40914 (best 4.40914), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
31/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' was not in top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
All Department Models Trained Successfully!
```

In [14]:

```python
# Create variables to help us calculate the overall average later
total_mape = 0
total_rmse = 0
num_departments = len(trained_predictors)

print("--- Department Evaluation Report ---")

# Loop through every department we trained
for dept in trained_predictors.keys():

    # 1. Grab the specific model and test data for THIS department
    current_predictor = trained_predictors[dept]
    current_test_ds = test_datasets[dept]

    # 2. This is YOUR exact code, just using the current department's data!
    forecast_it, ts_it = make_evaluation_predictions(
        dataset=current_test_ds,
        predictor=current_predictor,
        num_samples=100,
    )
```

```
    forecasts = list(forecast_it)
    tss = list(ts_it)

    evaluator = Evaluator(quantiles=[0.5])
    agg_metrics, item_metrics = evaluator(tss, forecasts)

    # 3. Print the results for this specific department
    print(f"Department {dept} | RMSE: {agg_metrics['RMSE']:.2f} | MAPE: {agg

    # Add to our running totals
    total_mape += agg_metrics['MAPE'] * 100
    total_rmse += agg_metrics['RMSE']

# 4. Calculate and print the grand totals for your presentation!
print("-" * 40)
print(f"OVERALL AVERAGE RMSE: {total_rmse / num_departments:.2f}")
print(f"OVERALL AVERAGE MAPE: {total_mape / num_departments:.2f}%")
print("-" * 40)
```

```
--- Department Evaluation Report ---
Running evaluation: 10it [00:00, 90.85it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_1 | RMSE: 63.27 | MAPE: 18.38%
Running evaluation: 10it [00:00, 119.23it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_2 | RMSE: 98.53 | MAPE: 9.98%
Running evaluation: 10it [00:00, 124.70it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_3 | RMSE: 195.37 | MAPE: 7.42%
Running evaluation: 10it [00:00, 114.66it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOBBIES_1 | RMSE: 61.01 | MAPE: 12.87%
Running evaluation: 10it [00:00, 89.81it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOBBIES_2 | RMSE: 11.89 | MAPE: 27.20%
```

```
Running evaluation: 10it [00:00, 123.37it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOUSEHOLD_1 | RMSE: 97.58 | MAPE: 8.60%
Running evaluation: 10it [00:00, 142.26it/s]
Department HOUSEHOLD_2 | RMSE: 28.32 | MAPE: 10.45%
----------------------------------------
OVERALL AVERAGE RMSE: 79.42
OVERALL AVERAGE MAPE: 13.56%
----------------------------------------
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
```

In [15]:

```
# 1. Pivot the price data
price_data = model_data.pivot_table(
    index='date',
    columns=['store_id', 'dept_id'],
    values='avg_price',
    aggfunc='mean'
)

# 2. Convert to datetime
price_data.index = pd.to_datetime(price_data.index)

# 3. Force continuous daily calendar (this adds NaNs for days with no sales
price_data = price_data.asfreq('D')

# 4. NOW fill the missing values
# 'ffill' carries yesterday's price forward for missing days.
# The .fillna(0) catches any items that had no price on the very first day.
price_data = price_data.fillna(method='ffill').fillna(0)
```

In [16]:

```
# 1. Get a list of all unique departments from your pivot table
departments = ts_data.columns.get_level_values('dept_id').unique()

# 2. Create dictionaries to store your trained Econ models and datasets
trained_econ_predictors = {}
test_econ_datasets = {}

prediction_length = 28
split_date = pd.to_datetime("2016-04-24")

# 3. Loop through each department and train a separate Econ model
for dept in departments:
    print(f"--- Preparing and Training Econ Model for Department: {dept} ---

    # Isolate SALES data for this department
    dept_ts_data = ts_data.xs(dept, level='dept_id', axis=1)
    train_dept_ts = dept_ts_data[dept_ts_data.index <= split_date]
```

```
    test_dept_ts = dept_ts_data

    # *** CHANGE 1: Isolate PRICE data for this department ***
    # (Assuming you already created 'price_data' from Cell 18)
    dept_price_data = price_data.xs(dept, level='dept_id', axis=1)
    train_dept_price = dept_price_data[dept_price_data.index <= split_date]
    test_dept_price = dept_price_data

    # *** CHANGE 2: Inject price into the Datasets ***
    train_ds_econ = ListDataset(
        [
            {
                "start": train_dept_ts.index[0],
                "target": train_dept_ts[col].values,
                "feat_dynamic_real": [train_dept_price[col].values]
            }
            for col in train_dept_ts.columns
        ],
        freq="D"
    )

    test_ds_econ = ListDataset(
        [
            {
                "start": test_dept_ts.index[0],
                "target": test_dept_ts[col].values,
                "feat_dynamic_real": [test_dept_price[col].values]
            }
            for col in test_dept_ts.columns
        ],
        freq="D"
    )

    # *** CHANGE 3: Update Estimator for dynamic features ***
    estimator_econ = DeepAREstimator(
        freq="D",
        prediction_length=prediction_length,
        context_length=prediction_length * 2,
        num_feat_dynamic_real=1,  # <--- Telling the AI to expect the price
        num_layers=2,
        hidden_size=32,
        trainer_kwargs={
            "max_epochs": 15,
            "accelerator": "cpu",
            "enable_progress_bar": False
        }
    )

    # Train the model
    predictor_econ = estimator_econ.train(train_ds_econ)

    # Save the trained predictor and test dataset
    trained_econ_predictors[dept] = predictor_econ
    test_econ_datasets[dept] = test_ds_econ

print("All Advanced Econ Department Models Trained Successfully!")
```

```
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
--------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
--------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: FOODS_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.03672 (best 6.03672), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_32/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.49688 (best 5.49688), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.43872 (best 5.43872), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.30663 (best 5.30663), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.24858 (best 5.24858), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.21927 (best 5.21927), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.18210 (best 5.18210), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' was not in top 1
Epoch 8, global step 450: 'train_loss' reached 5.14160 (best 5.14160), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
2/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' reached 5.13892 (best 5.13892), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
32/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' was not in top 1
Epoch 13, global step 700: 'train_loss' reached 5.13705 (best 5.13705), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
32/checkpoints/epoch=13-step=700.ckpt' as top 1
Epoch 14, global step 750: 'train_loss' reached 5.13562 (best 5.13562), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
32/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
```

```
--- Preparing and Training Econ Model for Department: FOODS_2 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.54843 (best 6.54843), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_33/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.86678 (best 5.86678), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.60667 (best 5.60667), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.52081 (best 5.52081), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.50828 (best 5.50828), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.49674 (best 5.49674), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.45038 (best 5.45038), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 5.42775 (best 5.42775), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.42252 (best 5.42252), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.41182 (best 5.41182), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
3/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 5.38747 (best 5.38747), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
33/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' reached 5.37880 (best 5.37880), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
33/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' reached 5.35801 (best 5.35801), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
33/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
--------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
--------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
```

```
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: FOODS_3 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 7.89817 (best 7.89817), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_34/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 7.13461 (best 7.13461), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 6.83057 (best 6.83057), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 6.78331 (best 6.78331), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 6.72803 (best 6.72803), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 6.70288 (best 6.70288), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 6.65626 (best 6.65626), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 6.65384 (best 6.65384), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' was not in top 1
Epoch 9, global step 500: 'train_loss' reached 6.63544 (best 6.63544), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
4/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 6.62556 (best 6.62556), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
34/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' reached 6.61875 (best 6.61875), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
34/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' reached 6.61333 (best 6.61333), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
34/checkpoints/epoch=13-step=700.ckpt' as top 1
Epoch 14, global step 750: 'train_loss' reached 6.60655 (best 6.60655), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
34/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
--------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
--------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
```

```
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: HOBBIES_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.31495 (best 6.31495), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_35/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 5.60210 (best 5.60210), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.55459 (best 5.55459), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.48008 (best 5.48008), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.45694 (best 5.45694), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.40845 (best 5.40845), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 5.39975 (best 5.39975), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 5.39892 (best 5.39892), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.38412 (best 5.38412), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
5/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' was not in top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' reached 5.37231 (best 5.37231), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
35/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: HOBBIES_2 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 4.12891 (best 4.12891), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_36/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 3.76045 (best 3.76045), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 3.72554 (best 3.72554), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 3.67453 (best 3.67453), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' was not in top 1
Epoch 5, global step 300: 'train_loss' reached 3.67166 (best 3.67166), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' reached 3.66334 (best 3.66334), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=6-step=350.ckpt' as top 1
Epoch 7, global step 400: 'train_loss' reached 3.65690 (best 3.65690), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 3.64999 (best 3.64999), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
6/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' was not in top 1
Epoch 11, global step 600: 'train_loss' was not in top 1
Epoch 12, global step 650: 'train_loss' reached 3.63472 (best 3.63472), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
36/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' was not in top 1
Epoch 14, global step 750: 'train_loss' reached 3.63397 (best 3.63397), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
36/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: HOUSEHOLD_1 ---
```

```
Epoch 0, global step 50: 'train_loss' reached 6.75874 (best 6.75874), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_37/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 6.18179 (best 6.18179), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 5.94011 (best 5.94011), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 5.83498 (best 5.83498), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 5.79608 (best 5.79608), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 5.72333 (best 5.72333), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' was not in top 1
Epoch 7, global step 400: 'train_loss' reached 5.68823 (best 5.68823), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=7-step=400.ckpt' as top 1
Epoch 8, global step 450: 'train_loss' reached 5.66835 (best 5.66835), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' reached 5.66301 (best 5.66301), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
7/checkpoints/epoch=9-step=500.ckpt' as top 1
Epoch 10, global step 550: 'train_loss' reached 5.65399 (best 5.65399), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
37/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' reached 5.64500 (best 5.64500), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
37/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' reached 5.63292 (best 5.63292), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
37/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' reached 5.61735 (best 5.61735), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
37/checkpoints/epoch=13-step=700.ckpt' as top 1
Epoch 14, global step 750: 'train_loss' was not in top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
GPU available: True (mps), used: False
TPU available: False, using: 0 TPU cores
HPU available: False, using: 0 HPUs

  | Name  | Type        | Params | Mode  | In sizes
| Out sizes
------------------------------------------------------------------------------
-----------------------------------------------------
0 | model | DeepARModel | 17.8 K | train | [[1, 1], [1, 1], [1, 1148, 5],
[1, 1148], [1, 1148], [1, 28, 5]] | [1, 100, 28]
------------------------------------------------------------------------------
-----------------------------------------------------
17.8 K    Trainable params
0         Non-trainable params
```

```
17.8 K    Total params
0.071     Total estimated model params size (MB)
11        Modules in train mode
0         Modules in eval mode
--- Preparing and Training Econ Model for Department: HOUSEHOLD_2 ---
Epoch 0, global step 50: 'train_loss' reached 5.44254 (best 5.44254), saving
model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_38/
checkpoints/epoch=0-step=50.ckpt' as top 1
Epoch 1, global step 100: 'train_loss' reached 4.87805 (best 4.87805), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=1-step=100.ckpt' as top 1
Epoch 2, global step 150: 'train_loss' reached 4.71577 (best 4.71577), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=2-step=150.ckpt' as top 1
Epoch 3, global step 200: 'train_loss' reached 4.52975 (best 4.52975), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=3-step=200.ckpt' as top 1
Epoch 4, global step 250: 'train_loss' reached 4.51503 (best 4.51503), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=4-step=250.ckpt' as top 1
Epoch 5, global step 300: 'train_loss' reached 4.44253 (best 4.44253), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=5-step=300.ckpt' as top 1
Epoch 6, global step 350: 'train_loss' was not in top 1
Epoch 7, global step 400: 'train_loss' was not in top 1
Epoch 8, global step 450: 'train_loss' reached 4.44009 (best 4.44009), savin
g model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_3
8/checkpoints/epoch=8-step=450.ckpt' as top 1
Epoch 9, global step 500: 'train_loss' was not in top 1
Epoch 10, global step 550: 'train_loss' reached 4.43980 (best 4.43980), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
38/checkpoints/epoch=10-step=550.ckpt' as top 1
Epoch 11, global step 600: 'train_loss' reached 4.43946 (best 4.43946), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
38/checkpoints/epoch=11-step=600.ckpt' as top 1
Epoch 12, global step 650: 'train_loss' reached 4.42874 (best 4.42874), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
38/checkpoints/epoch=12-step=650.ckpt' as top 1
Epoch 13, global step 700: 'train_loss' reached 4.41715 (best 4.41715), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
38/checkpoints/epoch=13-step=700.ckpt' as top 1
Epoch 14, global step 750: 'train_loss' reached 4.41365 (best 4.41365), savi
ng model to '/Users/conniezhang/Desktop/ECON491/data/lightning_logs/version_
38/checkpoints/epoch=14-step=750.ckpt' as top 1
`Trainer.fit` stopped: `max_epochs=15` reached.
All Advanced Econ Department Models Trained Successfully!
```

In [17]:

```python
# Create variables to track our overall averages for the Econ model
total_econ_mape = 0
total_econ_rmse = 0
num_departments = len(trained_econ_predictors)

print("--- Advanced (Econ) Department Evaluation Report ---")

# Loop through every Econ model we just trained
for dept in trained_econ_predictors.keys():
```

```
    # 1. Grab the specific Econ model and test data for THIS department
    current_predictor = trained_econ_predictors[dept]
    current_test_ds = test_econ_datasets[dept]

    # 2. Generate predictions
    forecast_it, ts_it = make_evaluation_predictions(
        dataset=current_test_ds,
        predictor=current_predictor,
        num_samples=100,
    )

    forecasts = list(forecast_it)
    tss = list(ts_it)

    evaluator = Evaluator(quantiles=[0.5])
    agg_metrics, item_metrics = evaluator(tss, forecasts)

    # 3. Print the results for this specific department
    print(f"Department {dept} | Econ RMSE: {agg_metrics['RMSE']:.2f} | Econ

    # Add to our running totals
    total_econ_mape += agg_metrics['MAPE'] * 100
    total_econ_rmse += agg_metrics['RMSE']

# 4. Calculate and print the grand totals!
print("-" * 40)
print(f"OVERALL ECON AVERAGE RMSE: {total_econ_rmse / num_departments:.2f}")
print(f"OVERALL ECON AVERAGE MAPE: {total_econ_mape / num_departments:.2f}%"
print("-" * 40)
```

```
--- Advanced (Econ) Department Evaluation Report ---
Running evaluation: 10it [00:00, 108.41it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_1 | Econ RMSE: 67.66 | Econ MAPE: 19.19%
Running evaluation: 10it [00:00, 104.24it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_2 | Econ RMSE: 93.83 | Econ MAPE: 10.54%
Running evaluation: 10it [00:00, 112.68it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department FOODS_3 | Econ RMSE: 194.84 | Econ MAPE: 7.62%
```

```
Running evaluation: 10it [00:00, 117.54it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOBBIES_1 | Econ RMSE: 59.36 | Econ MAPE: 12.74%
Running evaluation: 10it [00:00, 120.45it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOBBIES_2 | Econ RMSE: 12.09 | Econ MAPE: 26.07%
Running evaluation: 10it [00:00, 97.61it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Department HOUSEHOLD_1 | Econ RMSE: 91.72 | Econ MAPE: 8.31%
Running evaluation: 10it [00:00, 119.89it/s]
Department HOUSEHOLD_2 | Econ RMSE: 29.81 | Econ MAPE: 10.69%
----------------------------------------
OVERALL ECON AVERAGE RMSE: 78.47
OVERALL ECON AVERAGE MAPE: 13.59%
----------------------------------------
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
```

In [19]:

```python
from gluonts.evaluation import Evaluator

print("=" * 60)
print("FINAL OVERFITTING REPORT: TRAIN VS TEST".center(60))
print("=" * 60)

# Make sure we use the same split date as before
split_date = pd.to_datetime("2016-04-24")
evaluator = Evaluator(quantiles=[0.5])

for dept in departments:
    print(f"\n--- Analyzing Department: {dept} ---")

    # 1. Recreate the Training Data for this department (Base Model)
    dept_ts_data = ts_data.xs(dept, level='dept_id', axis=1)
    train_dept_ts = dept_ts_data[dept_ts_data.index <= split_date]
    train_ds_base = ListDataset(
        [{"start": train_dept_ts.index[0], "target": train_dept_ts[col].valu
        freq="D"
    )

    # 2. Recreate the Training Data for this department (Econ Model with Pri
```

```
    dept_price_data = price_data.xs(dept, level='dept_id', axis=1)
    train_dept_price = dept_price_data[dept_price_data.index <= split_date]
    train_ds_econ = ListDataset(
        [
            {
                "start": train_dept_ts.index[0],
                "target": train_dept_ts[col].values,
                "feat_dynamic_real": [train_dept_price[col].values]
            }
            for col in train_dept_ts.columns
        ],
        freq="D"
    )

    # --- EVALUATE BASE MODEL (TRAIN VS TEST) ---
    # Train
    fcst_train_base, ts_train_base = make_evaluation_predictions(dataset=tra
    metrics_train_base, _ = evaluator(list(ts_train_base), list(fcst_train_b

    # Test (using the test datasets we saved earlier)
    fcst_test_base, ts_test_base = make_evaluation_predictions(dataset=test_
    metrics_test_base, _ = evaluator(list(ts_test_base), list(fcst_test_base

    # --- EVALUATE ECON MODEL (TRAIN VS TEST) ---
    # Train
    fcst_train_econ, ts_train_econ = make_evaluation_predictions(dataset=tra
    metrics_train_econ, _ = evaluator(list(ts_train_econ), list(fcst_train_e

    # Test
    fcst_test_econ, ts_test_econ = make_evaluation_predictions(dataset=test_
    metrics_test_econ, _ = evaluator(list(ts_test_econ), list(fcst_test_econ

    # --- PRINT THE RESULTS ---
    print("  [Base DeepAR]")
    print(f"    Train MAPE: {metrics_train_base['MAPE']*100:.2f}%  |  Test M

    print("  [Advanced Econ DeepAR (w/ Price)]")
    print(f"    Train MAPE: {metrics_train_econ['MAPE']*100:.2f}%  |  Test M

print("\n" + "=" * 60)
print("REPORT COMPLETE")
```

```
============================================================
          FINAL OVERFITTING REPORT: TRAIN VS TEST
============================================================

--- Analyzing Department: FOODS_1 ---
```

```
Running evaluation: 10it [00:00, 109.05it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 154.72it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 68.51it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 95.64it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
  [Base DeepAR]
    Train MAPE: 17.90%  |  Test MAPE: 18.47%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 18.74%  |  Test MAPE: 19.00%

--- Analyzing Department: FOODS_2 ---
Running evaluation: 10it [00:00, 99.04it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 147.87it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 143.24it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 141.77it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
```

```
  [Base DeepAR]
    Train MAPE: 10.13%  |  Test MAPE: 10.09%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 10.56%  |  Test MAPE: 10.53%

--- Analyzing Department: FOODS_3 ---
Running evaluation: 10it [00:00, 149.03it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 87.64it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 88.89it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 106.80it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
  [Base DeepAR]
    Train MAPE: 7.40%  |  Test MAPE: 7.39%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 7.61%  |  Test MAPE: 7.69%

--- Analyzing Department: HOBBIES_1 ---
```

```
Running evaluation: 10it [00:00, 103.71it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 140.36it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 144.07it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 128.03it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
  [Base DeepAR]
    Train MAPE: 12.84%  |  Test MAPE: 12.69%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 12.46%  |  Test MAPE: 12.78%

--- Analyzing Department: HOBBIES_2 ---
Running evaluation: 10it [00:00, 122.82it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 61.34it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 119.48it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 122.51it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
```

```
  [Base DeepAR]
    Train MAPE: 27.14%  |  Test MAPE: 26.95%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 26.80%  |  Test MAPE: 26.61%

--- Analyzing Department: HOUSEHOLD_1 ---
Running evaluation: 10it [00:00, 102.36it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 148.62it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 147.36it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 115.77it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
  [Base DeepAR]
    Train MAPE: 8.53%  |  Test MAPE: 8.74%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 8.31%  |  Test MAPE: 8.35%

--- Analyzing Department: HOUSEHOLD_2 ---
Running evaluation: 10it [00:00, 121.80it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 126.93it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 93.60it/s]
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
Running evaluation: 10it [00:00, 111.72it/s]
```

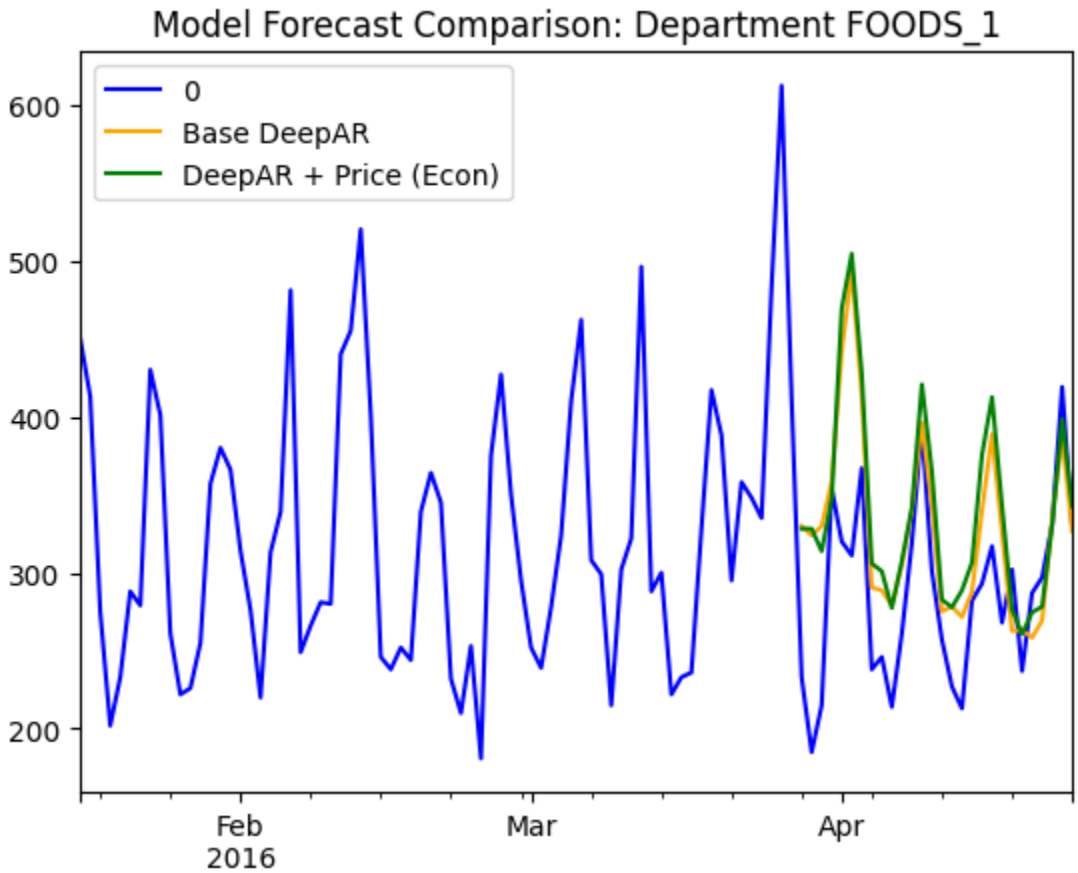
```
  [Base DeepAR]
    Train MAPE: 10.54%  |  Test MAPE: 10.51%
  [Advanced Econ DeepAR (w/ Price)]
    Train MAPE: 10.76%  |  Test MAPE: 10.77%

============================================================
REPORT COMPLETE
/opt/miniconda3/envs/econ491_walmart_forcasting/lib/python3.10/site-package
s/gluonts/json.py:102: UserWarning: Using `json`-module for json-handling. C
onsider installing one of `orjson`, `ujson` to speed up serialization and de
serialization.
  warnings.warn(
```

In [20]:

```
import matplotlib.pyplot as plt

# 1. Choose which department you want to visualize for your presentation
target_dept = "FOODS_1"  # Change this to 'HOBBIES_1', 'FOODS_2', etc., to s

print(f"Generating plot for {target_dept}...")

# 2. Grab predictions for the Base Model
forecast_it_base, ts_it_base = make_evaluation_predictions(
    dataset=test_datasets[target_dept],
    predictor=trained_predictors[target_dept],
    num_samples=100,
)
forecasts_base = list(forecast_it_base)
tss_base = list(ts_it_base)

# 3. Grab predictions for the Advanced (Econ) Model
forecast_it_econ, ts_it_econ = make_evaluation_predictions(
    dataset=test_econ_datasets[target_dept],
    predictor=trained_econ_predictors[target_dept],
    num_samples=100,
)
forecasts_econ = list(forecast_it_econ)

# 4. Extract the first store's time series from this specific department
ts = tss_base[0]
forecast_base = forecasts_base[0]
forecast_econ = forecasts_econ[0]

# 5. Plot the results!
plt.figure(figsize=(12,6))

# Actual values (looking at the last 100 days so we can see the forecast cle
ts[-100:].plot(color='blue', label="Actual Sales")

# Base model prediction (median)
plt.plot(forecast_base.index, forecast_base.quantile(0.5), color='orange', l

# Econ model prediction (median)
plt.plot(forecast_econ.index, forecast_econ.quantile(0.5), color='green', la

plt.title(f"Model Forecast Comparison: Department {target_dept}")
plt.legend()
```

```
plt.show()
```

```
Generating plot for FOODS_1...
<Figure size 1200x600 with 0 Axes>
```



In [ ]: